

# **SEMICONDUCTOR YIELD PREDICTION USING MACHINE LEARNING**

**Project Report**

Submitted by

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## Abstract

Semiconductor manufacturing involves several process parameters that can affect the final product yield. Predicting semiconductor yield can help identify low-yield conditions and improve manufacturing quality.

This project develops a machine learning-based semiconductor yield prediction system using a semiconductor manufacturing dataset. The data is preprocessed and standardized before training different classification algorithms. Random Forest, Support Vector Machine (SVM), and Logistic Regression models are trained and evaluated.

Among the evaluated models, the Support Vector Machine (SVM) achieved the highest accuracy of approximately 93.63%. The trained SVM model is saved using Joblib and used for predicting new semiconductor samples.

The developed system successfully classifies semiconductor samples into High Yield and Low Yield categories and provides prediction probabilities. This demonstrates the usefulness of machine learning for semiconductor manufacturing yield analysis.

## Keywords

Semiconductor Manufacturing, Yield Prediction, Machine Learning, Classification, Random Forest, Support Vector Machine (SVM), Logistic Regression, Feature Standardization.

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# 1. INTRODUCTION

Semiconductor manufacturing is a complex process that involves various process and sensor parameters. These parameters can significantly affect the quality and yield of the final semiconductor products. Therefore, predicting semiconductor yield is important for identifying potential low-yield conditions and improving manufacturing quality.

Machine learning techniques can be used to analyze manufacturing data and identify patterns that are difficult to detect using traditional methods. In this project, machine learning classification algorithms are applied to semiconductor manufacturing data to predict whether a semiconductor sample belongs to the High Yield or Low Yield category.

The project uses a semiconductor manufacturing dataset containing multiple process and sensor-related features. The dataset is preprocessed and standardized before applying machine learning algorithms. Three classification models, namely Random Forest, Support Vector Machine (SVM), and Logistic Regression, are trained and evaluated.

The performance of the models is compared using classification evaluation metrics. Among the evaluated models, the Support Vector Machine achieved the highest accuracy of approximately 93.63% and was selected as the best model for prediction.

The trained model is saved using Joblib and used to predict new semiconductor samples. The system also provides prediction probabilities for High Yield and Low Yield categories. This project demonstrates how machine learning can support semiconductor manufacturing quality analysis and yield prediction.

## 1.1 Problem Statement

Semiconductor manufacturing involves numerous process parameters, and variations in these parameters can affect product yield. Identifying low-yield conditions manually can be difficult when the dataset contains a large number of sensors and process features. Therefore, there is a need for a machine learning-based system that can analyze manufacturing data and predict semiconductor yield accurately.

## 1.2 Objectives

- To analyze semiconductor manufacturing data.
- To preprocess and standardize the dataset.
- To train different machine learning classification models.
- To compare the performance of Random Forest, SVM, and Logistic Regression.
- To identify the best-performing model.
- To predict High Yield and Low Yield semiconductor samples.
- To provide prediction probabilities for new samples.

## 2. DATASET DESCRIPTION

### 2.1 Dataset Overview

The dataset used in this project contains semiconductor manufacturing process and sensor-related data. The dataset is used to analyze manufacturing conditions and predict semiconductor yield.

The dataset contains multiple input features representing different manufacturing and sensor parameters. A target variable is used to classify the semiconductor samples into High Yield and Low Yield categories.

The dataset is stored in CSV format and was loaded into the Python environment using the Pandas library. The dataset was examined to understand its structure, features, and target variables before applying machine learning techniques.

### 2.2 Dataset Loading

The semiconductor dataset was loaded using the Pandas `read_csv()` function. The dataset was then inspected using functions such as `head()`, `info()`, and descriptive statistics to understand the available data.

| Time  | A       | B       | C        | D        | E      | F   | G        | H        | I      | J       | K       | L       | M        | N        | O       | P        | Q        | R       | S        | T        | U       | V        | W        |
|-------|---------|---------|----------|----------|--------|-----|----------|----------|--------|---------|---------|---------|----------|----------|---------|----------|----------|---------|----------|----------|---------|----------|----------|
|       | 0       | 1       | 2        | 3        | 4      | 5   | 6        | 7        | 8      | 9       | 10      | 11      | 12       | 13       | 14      | 15       | 16       | 17      | 18       | 19       | 20      | 21       |          |
| ##### | 3030.93 | 2564    | 2187.733 | 1411.127 | 1.3602 | 100 | 97.6133  | 0.1242   | 1.5005 | 0.0162  | -0.0034 | 0.9455  | 202.4396 | 0        | 7.9558  | 414.871  | 10.0433  | 0.968   | 192.3963 | 12.519   | 1.4026  | -5419    |          |
| ##### | 3095.78 | 2465.14 | 2230.422 | 1463.661 | 0.8294 | 100 | 102.3433 | 0.1247   | 1.4966 | -0.0005 | -0.0148 | 0.9627  | 200.547  | 0        | 10.1548 | 414.7347 | 9.2599   | 0.9701  | 191.2872 | 12.4608  | 1.3825  | -5441.5  |          |
| ##### | 2932.61 | 2559.94 | 2186.411 | 1698.017 | 1.5102 | 100 | 95.4878  | 0.1241   | 1.4436 | 0.0041  | 0.0013  | 0.9615  | 202.0179 | 0        | 9.5157  | 416.7075 | 9.3144   | 0.9674  | 192.7035 | 12.5404  | 1.4123  | -5447.75 |          |
| ##### | 2988.72 | 2479.9  | 2199.033 | 909.7926 | 1.3204 | 100 | 104.2367 | 0.1217   | 1.4882 | -0.0124 | -0.0033 | 0.9629  | 201.8482 | 0        | 9.6052  | 422.2894 | 9.6924   | 0.9687  | 192.1557 | 12.4782  | 1.4011  | -5468.25 |          |
| ##### | 3032.24 | 2502.87 | 2233.367 | 1326.52  | 1.5334 | 100 | 100.3967 | 0.1235   | 1.5031 | -0.0031 | -0.0072 | 0.9569  | 201.9424 | 0        | 10.5661 | 420.5925 | 10.3387  | 0.9735  | 191.6037 | 12.4735  | 1.3888  | -5476.25 |          |
| ##### | 2946.25 | 2432.84 | 2233.367 | 1326.52  | 1.5334 | 100 | 100.3967 | 0.1235   | 1.5287 | 0.0167  | 0.0055  | 0.9699  | 200.472  | 0        | 8.6617  | 414.2426 | 9.2441   | 0.9747  | 191.228  | 12.4935  | 1.3836  | -6058    |          |
| ##### | 3030.27 | 2430.12 | 2230.422 | 1463.661 | 0.8294 | 100 | 102.3433 | 0.1247   | 1.5816 | -0.027  | 0.0105  | 0.9591  | 202.0901 | 0        | 9.035   | 415.8852 | 9.999    | 0.9667  | 192.0912 | 12.4608  | 1.4067  | -6154    |          |
| ##### | 3058.88 | 2690.15 | 2248.9   | 1004.469 | 0.7884 | 100 | 106.24   | 0.1185   | 1.5153 | 0.0157  | 0.0007  | 0.9481  | 202.417  | 0        | 13.6872 | 408.4017 | 9.6836   | 0.9687  | 192.7334 | 12.5263  | 1.4092  | -5395.5  |          |
| ##### | 2967.68 | 2600.47 | 2248.9   | 1004.469 | 0.7884 | 100 | 106.24   | 0.1185   | 1.5358 | 0.0111  | -0.0066 | 0.9494  | 202.4544 | 0        | 12.6837 | 417.6009 | 9.7046   | 0.9693  | 192.7498 | 12.5263  | 1.4278  | -4196.5  |          |
| ##### | 3016.11 | 2428.37 | 2248.9   | 1004.469 | 0.7884 | 100 | 106.24   | 0.1185   | 1.5381 | 0.0159  | 0.0049  | 0.944   | 202.5999 | 0        | 12.4278 | 413.3677 | 9.7046   | 0.9667  | 192.8953 | 12.5263  | 1.4185  | -5357.5  |          |
| ##### | 2994.05 | 2548.21 | 2195.122 | 1046.147 | 1.3204 | 100 | 103.34   | 0.1223   | 1.5144 | -0.019  | 0.0013  | 0.9433  | 201.7125 | 0        | 11.8566 | 411.9572 | 10.2918  | 0.9664  | 191.4207 | 12.4663  | 1.4195  | -2727.5  |          |
| ##### | 2928.84 | 2479.4  | 2196.211 | 1605.758 | 0.9959 | 100 | 97.9156  | 0.1257   | 1.469  | 0.017   | -0.0154 | 0.9445  | 202.1264 | 0        | 9.1084  | 419.9018 | 10.113   | 0.9673  | 192.0134 | 12.5263  | 1.421   | -6097.75 |          |
| ##### | 2920.07 | 2507.4  | 2195.122 | 1046.147 | 1.3204 | 100 | 103.34   | 0.1223   | 1.531  | -0.0259 | 0.0216  | 0.9595  | 202.1269 | 0        | 8.4828  | 415.5185 | 9.5007   | 0.9666  | 192.6261 | 12.4186  | 1.3848  | -5521.25 |          |
| ##### | 3051.44 | 2529.27 | 2184.433 | 877.6266 | 1.4668 | 100 | 107.8711 | 0.124    | 1.5236 | -0.0209 | -0.0031 | 0.9441  | 226.0086 | 0        | 9.7686  | 409.8885 | 10.4109  | 0.9516  | 215.5977 | 10.7029  | 1.4191  | -6069.75 |          |
| ##### | 2963.97 | 2629.48 | 2224.622 | 947.7739 | 1.2924 | 100 | 104.8489 | 0.1197   | 1.4474 | 0.0144  | -0.0119 | 0.9582  | 195.3787 | 0        | 9.7561  | 422.3675 | 9.7825   | 0.9692  | 185.5963 | 12.4582  | 1.4029  | -1431.5  |          |
| ##### | 2988.31 | 2546.26 | 2224.622 | 947.7739 | 1.2924 | 100 | 104.8489 | 0.1197   | 1.5465 | 0.025   | -0.0024 | 0.9616  | 192.9787 | 0        | 12.4364 | 424.7536 | 9.5917   | 0.9706  | 183.3869 | 12.4622  | 1.4088  | -2694.75 |          |
| ##### | 3028.02 | 2560.87 | 2270.256 | 1258.456 | 1.395  | 100 | 104.8078 | 0.1207   | 1.4368 | 0.015   | -0.0037 | 0.9623  | 195.1742 | 0        | 12.1805 | 428.9826 | 9.1999   | 0.9673  | 185.9743 | 12.4761  | 1.3969  | -5757.25 |          |
| ##### | 3032.73 | 2517.79 | 2270.256 | 1258.456 | 1.395  | 100 | 104.8078 | 0.1207   | 1.5537 | 0.022   | -0.0027 | 0.9613  | 195.3425 | 0        | 10.0002 | 420.4726 | 9.4147   | 0.9692  | 185.9278 | 12.4761  | 1.403   | -5612.5  |          |
| ##### | 3040.34 | 2501.16 | 2207.389 | 962.5317 | 1.2043 | 100 | 104.0311 | 0.121    | 1.5481 | -0.0367 | 0.0014  | 0.9634  | 196.2746 | 0        | 8.4061  | 409.1399 | 9.847    | 0.9701  | 186.4275 | 12.5222  | 1.4208  | -5575.25 |          |
| ##### | 2988.3  | 2519.05 | 2208.856 | 1157.722 | 1.5509 | 100 | 107.8022 | 0.1233   | 1.5362 | -0.0259 | -0.0179 | 0.9614  | 197.1793 | 0        | 13.3419 | 404.5667 | 10.1924  | 0.9697  | 186.9869 | 12.5635  | 1.4075  | -5469.25 |          |
| ##### | 2987.32 | 2528.81 |          |          |        |     |          | 0.1195   | 1.6343 | -0.0263 | 0.0116  | 0.9587  | 200.8256 | 0        | 11.9224 | 414.306  | 9.7659   | 0.9661  | 191.0598 | 12.4761  | 1.4014  | -5441    |          |
| ##### |         | 2481.85 | 2207.389 | 962.5317 | 1.2043 |     | 100      | 104.0311 | 0.121  | 1.5559  | 0.0002  | -0.0044 | 0.9617   | 196.6315 | 0       | 13.6262  | 410.695  | 10.3503 | 0.9684   | 186.2811 | 12.4761 | 1.4061   | -5434    |
| ##### | 3002.27 | 2497.45 | 2207.389 | 962.5317 | 1.2043 |     | 100      | 104.0311 | 0.121  | 1.5465  | 0.0195  | -0.0114 | 0.9491   | 199.6394 | 0       | 15.4346  | 418.2477 | 10.3647 | 0.9706   | 189.2748 | 12.4761 | 1.406    | -5310.25 |

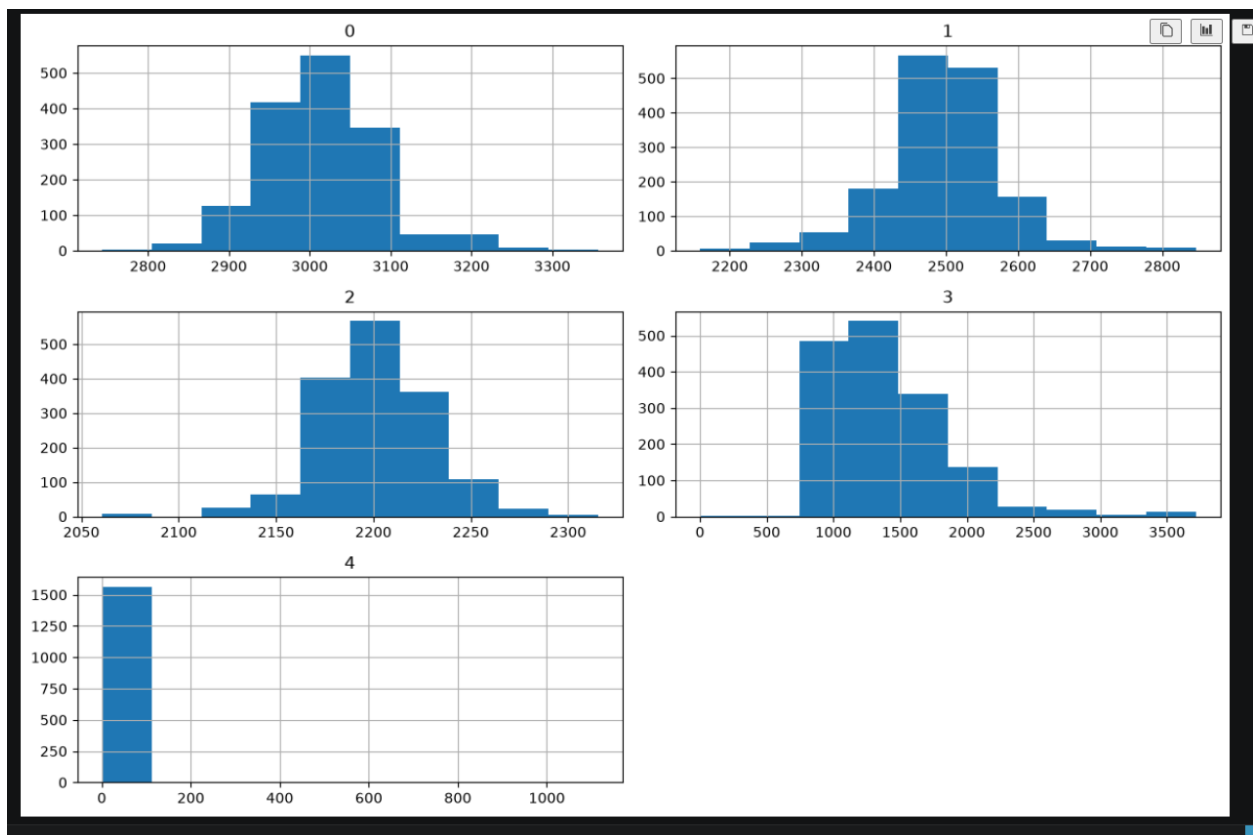
## 3. EXPLORATORY DATA ANALYSIS

### 3.1 Overview

Exploratory Data Analysis (EDA) was performed to understand the distribution of semiconductor manufacturing data and the relationships between different features.

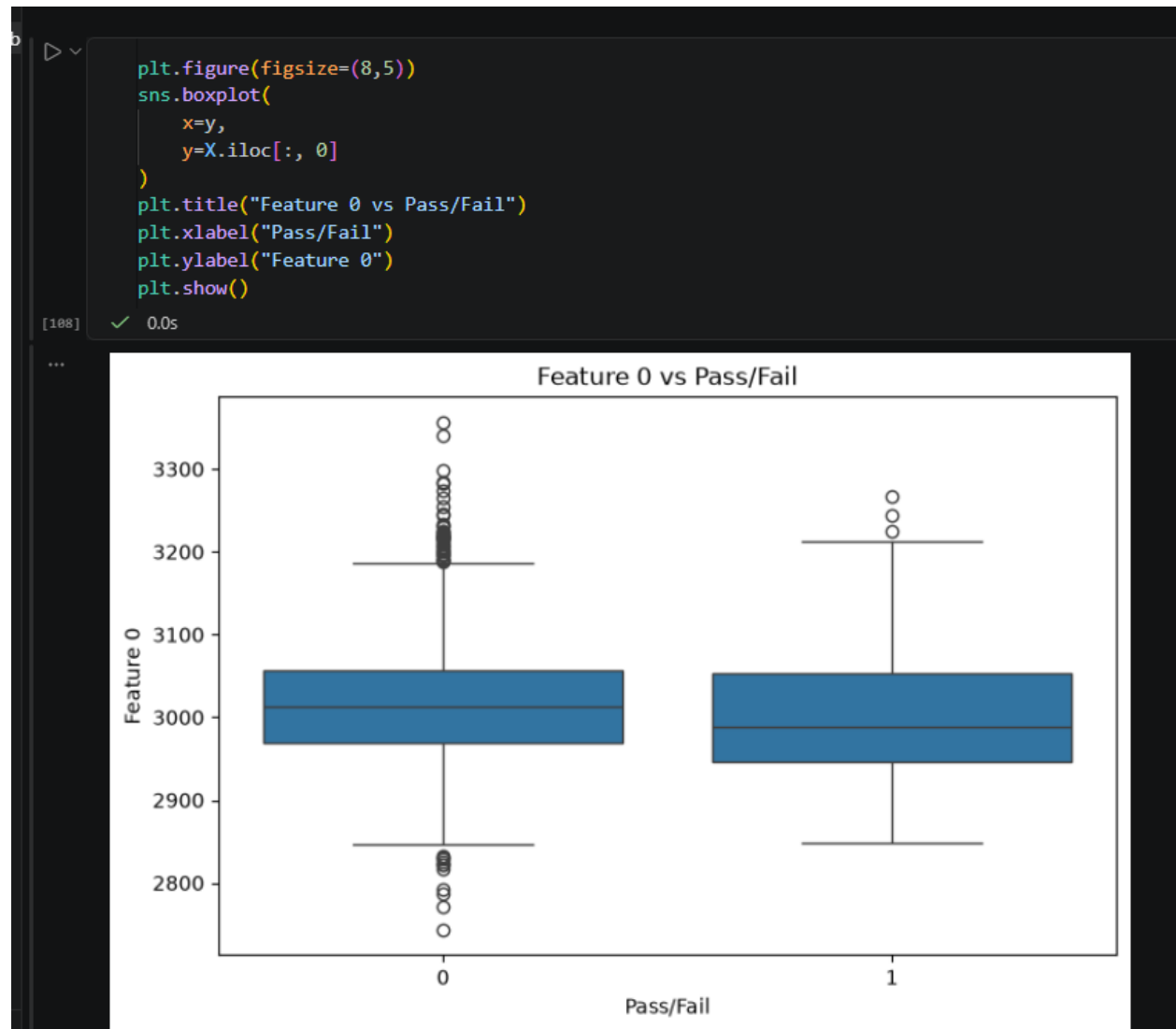
### 3.2 Univariate Analysis

Univariate analysis was performed to study the distribution of individual sensors and process features. Histograms were used to visualize the distribution and variation of selected features.



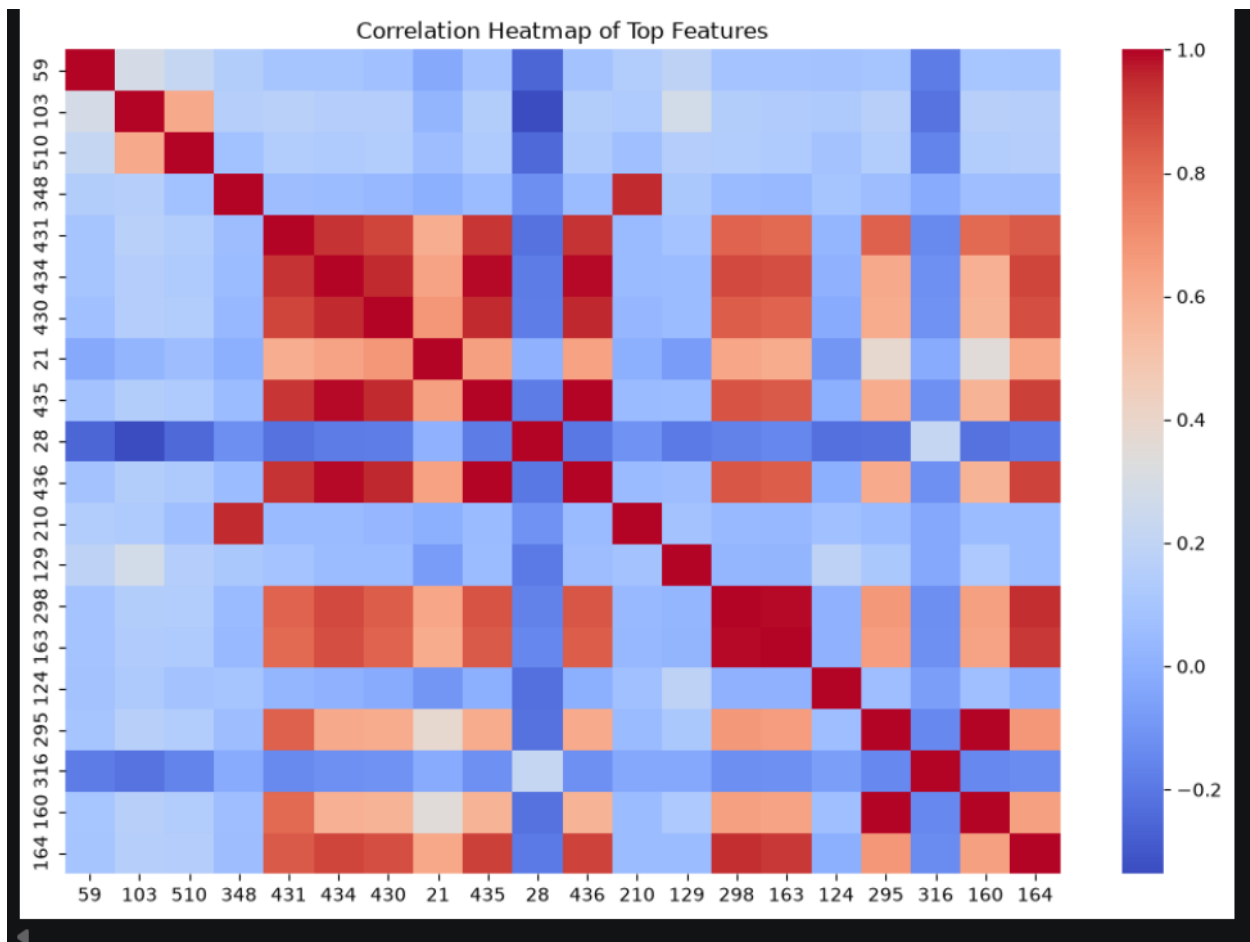
### 3.3 Bivariate Analysis

Bivariate analysis was performed to study the relationship between individual features and the semiconductor yield category. Box plots were used to compare feature distributions across different yield classes.



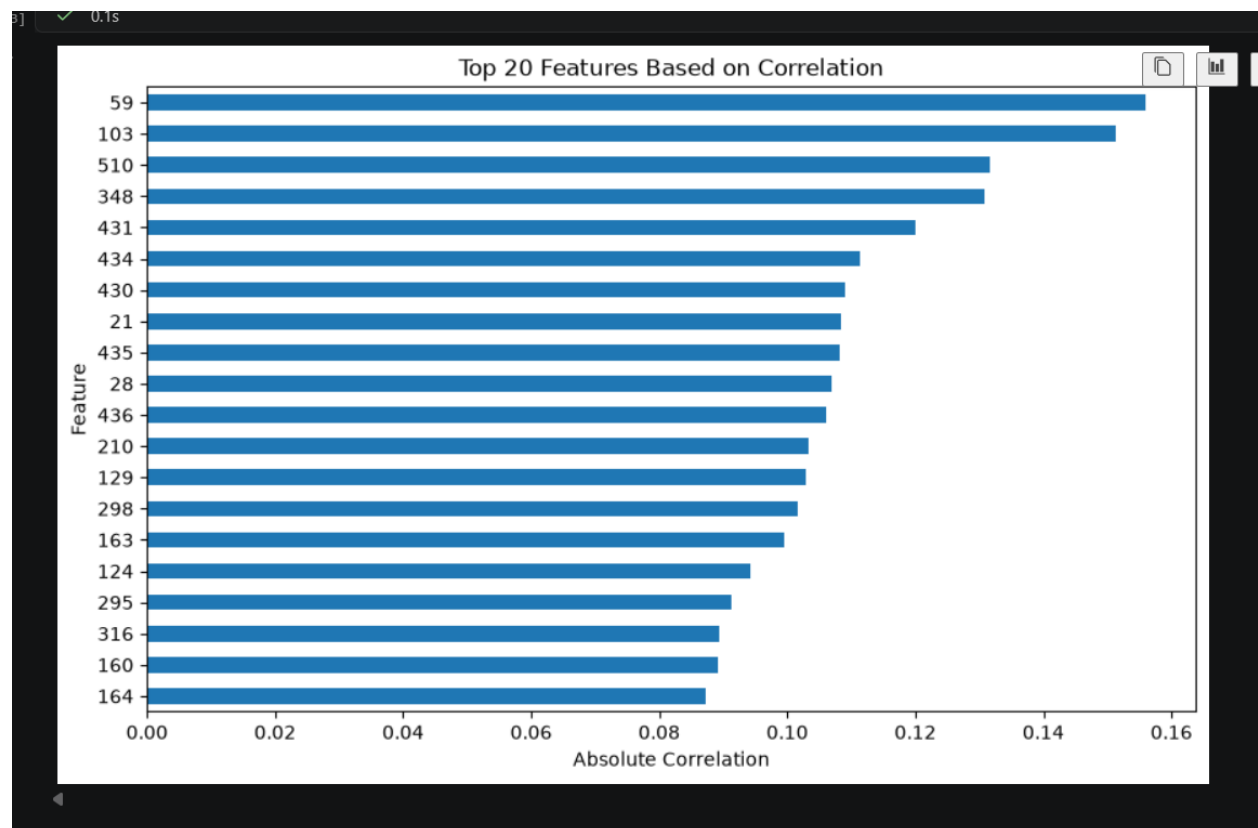
### 3.4 Multivariate Analysis

Multivariate analysis was performed using correlation analysis. A correlation heatmap was used to visualize the relationships among important features.



### 3.5 Feature Importance

Feature importance analysis was performed to identify the features that have a significant relationship with semiconductor yield prediction. The Random Forest model was used to analyze feature importance.



## 4. DATA PROCESSING

### 4.1 Data Cleaning

The dataset was checked for missing values and inconsistent data. Appropriate preprocessing techniques were applied to prepare the dataset for machine learning.

### 4.2 Feature and Target Selection

The input features were separated from the target variable. The input features represent semiconductor manufacturing parameters, while the target variable represents the yield classification.

### 4.3 Data Splitting

The dataset was divided into training and testing sets. The training data was used to train the machine learning models, while the testing data was used to evaluate their performance.

### 4.4 Feature Scaling

Feature scaling was performed using StandardScaler. Scaling ensures that features with different numerical ranges contribute appropriately to the machine learning models.

```
scaler= StandardScaler()  
X_train_scaled= scaler.fit_transform(X_train)  
X_test_scaled= scaler.transform(X_test)
```

✓ 0.0s

## 5. MACHINE LEARNING MODELS

### 5.1 Random Forest

Random Forest is an ensemble machine learning classification algorithm that combines multiple decision trees to produce a final prediction. In this project, Random Forest was trained using the processed semiconductor manufacturing dataset.

### 5.2 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification. It identifies an optimal decision boundary between different classes. In this project, SVM was trained to classify semiconductor samples into High Yield and Low Yield categories.

### 5.3 Logistic Regression

Logistic Regression is a classification algorithm used to predict the probability of a sample belonging to a particular class. It was also trained and evaluated for semiconductor yield prediction.

### 5.4 Model Comparison

The three machine learning models were trained and evaluated using the same processed dataset. Their performance was compared using classification metrics such as accuracy, precision, recall, and F1-score.

Among the evaluated models, **Support Vector Machine (SVM) achieved the highest accuracy of approximately 93.63%** and was selected as the best-performing model.

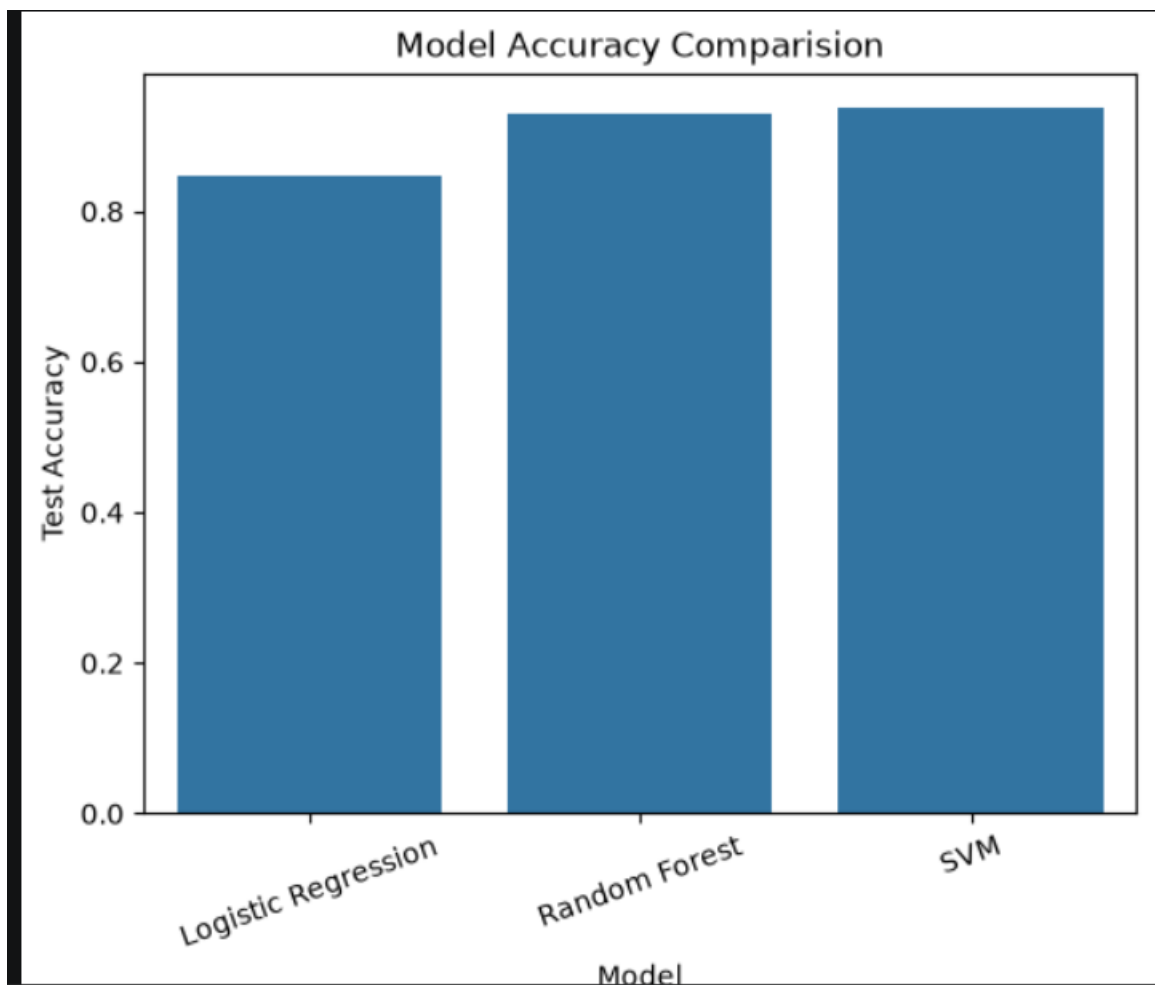
## 6. MODEL COMPARISON

Three machine learning classification algorithms were evaluated for semiconductor yield prediction:

1. Random Forest
2. Support Vector Machine (SVM)
3. Logistic Regression

The performance of the models was compared using accuracy and other classification evaluation metrics.

Among the evaluated models, **Support Vector Machine (SVM)** achieved the **highest accuracy of approximately 93.63%**. Therefore, SVM was selected as the best-performing model for semiconductor yield prediction.



## 7. PREDICTION RESULTS

The trained Support Vector Machine model was used to predict the yield category of a new semiconductor sample. The prediction result was:

**Prediction Class:** 0

**Result:** Low Yield Semiconductor

**Low Yield Probability:** 99.9997%

**High Yield Probability:** 0.0003%

**Model Accuracy:** 93.63%

The result indicates that the given sample has a very high probability of belonging to the **Low Yield Semiconductor** category. The trained SVM model and scaler were saved using Joblib and successfully loaded for prediction.

```
print("=====")
print("SEMICONDUCTOR YIELD PREDICTION")
print("=====")
print("Selected Model: SVM")
print("Model Accuracy:", accuracy_score(y_test, y_pred_svm))
print("Prediction:", prediction[0])
if prediction[0] == 1:
    print("Result: High Yield Semiconductor")
else:
    print("Result: Low Yield Semiconductor")
print("Low Yield Probability:", probability[0][0])
print("High Yield Probability:", probability[0][1])
print("=====")
print("Prediction completed successfully!")
print("=====")
```

✓ 0.0s

```
=====
SEMICONDUCTOR YIELD PREDICTION
=====
Selected Model: SVM
Model Accuracy: 0.9363057324840764
Prediction: 0
Result: Low Yield Semiconductor
Low Yield Probability: 0.9999967249616706
High Yield Probability: 3.2750383294426992e-06
=====
Prediction completed successfully!
=====
```

## 8. CONCLUSION

This project developed a machine learning-based system for predicting semiconductor manufacturing yield.

Three classification algorithms, namely Random Forest, Support Vector Machine (SVM), and Logistic Regression, were trained and evaluated. Among these models, **SVM achieved the highest accuracy of approximately 93.63%.**

The trained SVM model was saved using Joblib and successfully used to predict new semiconductor samples. The system classified samples into High Yield and Low Yield categories and also provided prediction probabilities.

The results demonstrate that machine learning can be effectively applied to semiconductor manufacturing data for yield prediction and quality analysis.

## 9. FUTURE ENHANCEMENT

The project can be further improved by:

- Testing additional machine learning algorithms.
- Performing hyperparameter optimization.
- Using a larger and more diverse semiconductor dataset.
- Developing a web-based prediction interface.
- Integrating real-time manufacturing sensor data.
- Applying Explainable AI techniques to understand model predictions.

## 10. REFERENCES

1. Pandas Documentation – Data analysis and manipulation using Python.
2. Scikit-learn Documentation – Machine learning algorithms and evaluation techniques.
3. Joblib Documentation – Model saving and loading in Python.
4. Python Documentation – Python programming language.

